

Vinoth M

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PROFESSIONAL SUMMARY

Associate Software Engineer with 2+ years building scalable distributed systems and high-throughput enterprise API integrations. Proficient in Java, Python, and Node.js with hands-on experience in system design (HLD/LLD), SOLID principles, design patterns, Redis caching, AWS SQS, and 12-Factor app principles. Delivered backend solutions reducing manual effort by 60–70% across production systems.

EXPERIENCE

Associate Software Engineer | Aptean | Coimbatore, India

Sep 2024 – Present

- **Distributed Systems & API Design:** Collaborated with architects to define data flow specs and integration standards for 10+ carrier APIs (incl. FedEx) across a B2B logistics platform serving 10K+ daily requests.
- **LLD & Design Patterns:** Designed carrier mapping engine applying SOLID principles, Factory pattern for carrier instantiation, and Strategy pattern for pluggable request/response mapping, enabling zero-code carrier onboarding and eliminating 40 hrs/month of manual configuration.
- **Redis Caching:** Implemented Redis caching for carrier-specific code lookups, eliminating repeated DB hits per request and reducing average response latency across request/response mapping workflows.
- **12-Factor App & Observability:** Applied 12-Factor methodology across backend services: env-based config, stateless service design, backing service abstraction (Redis, PostgreSQL). Monitored API latency and error rates via CloudWatch; traced and resolved 20+ production issues across Java, .NET Core, Node.js, and React.
- **Java Backend & Concurrency:** Developed Java-based carrier integration services using OOP design patterns and multithreading for concurrent request handling. Optimized React frontend via code-splitting, improving Time-to-Interactive by 40%.
- **RAG System:** Architected production RAG system using Azure OpenAI and pgvector (PostgreSQL) with semantic vector search, enabling context-aware carrier documentation queries and reducing engineer lookup time by 60%.
- **Multi-Agent AI Workflows:** Engineered multi-agent workflows (Azure OpenAI) with Chain-of-Thought and Few-Shot prompt engineering; agents parse specs, generate mapping configs, and trigger validation, cutting manual onboarding effort by 70%.
- **Database & Schema Design:** Managed PostgreSQL and SQL Server workflows: schema design, query optimization, and data pipelines at scale. Guided junior developers and authored design documentation across teams.
- **Cross-functional Collaboration:** Facilitated collaboration across distributed teams (India & China) in an Agile one-product model; drove code reviews across 10+ members; delivered sprint reviews to leads and business stakeholders.

PROJECTS

DocReady: Document Analysis & Migration Readiness Tool [\[GitHub\]](#) Python, Gemini AI, React, AWS SQS, REST API, .docx/.pdf

- **Async Job Queue:** Implemented AWS SQS to asynchronously queue document analysis jobs, decoupling upload handling from LLM processing pipeline and enabling concurrent multi-document processing.
- Built LLM-powered verdict engine producing structured readiness scores (READY → NOT READY) with actionable strengths, issues, and improvement suggestions per document.

JobNest: Microservices Job Portal [\[GitHub\]](#) Spring Boot (Java), MySQL, Redis, Redux, JWT, RESTful APIs

- Engineered scalable job portal with Spring Boot microservices (auth, jobs, applications) secured with JWT and RBAC; normalized MySQL schema with indexed queries reducing search latency by 35%.
- Built responsive React + Redux frontend with role-based views for job seekers and employers, consuming RESTful APIs across independently deployable microservices.

LinkedIn Content Agent [\[GitHub\]](#) Node.js, Gemini API, Multi-Agent AI, Prompt Engineering, React, OAuth 2.0

- Built full-stack AI automation platform using a 3-agent pipeline (Planner, Writer, Reviewer) with Chain-of-Thought prompting; integrated LinkedIn OAuth 2.0 + publishing API with live agent-reasoning traces in React frontend.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C#

Backend: Spring Boot, Node.js, .NET Core, Express.js, RESTful APIs, Microservices, WebSocket

Databases & Cache: PostgreSQL, MySQL, SQL Server, Redis (Caching), pgvector (Vector Search)

Cloud & Infra: AWS (Lambda, S3, SQS, API Gateway, CloudWatch, RDS), Docker, 12-Factor App

AI / LLM: Azure OpenAI, OpenAI API, RAG Systems, Multi-Agent AI, Prompt Engineering, Embeddings, Vector Search

Concepts: System Design (HLD/LLD), Distributed Systems, SOLID, Design Patterns (Factory, Strategy), Multithreading, DSA, JWT, OAuth, Agile

EDUCATION

B.Tech in Information Technology | Sri Krishna College of Engineering and Technology, Coimbatore Dec 2021 – May 2025 | CGPA: 7.75/10

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon 2023: Qualified internal round, outperforming 150+ teams. | LeetCode: 400+ problems solved across DSA and system design.
- Certifications: AWS Academy Cloud Foundations | Red Hat OpenShift | Oracle Java SE 8 | Coursera Python | Udemy React & Redux